AI

AI for Research

UTM BIG DATA CENTRE

MODULE 4 OF 5 · PRINTABLE NOTES

Data and Research Visualisation

Interpret research findings, select appropriate visual formats and produce diagrams, infographics, charts or data narratives that communicate results clearly.

Chart-Type Helper | Bibliometric Mapping | Honest Axes | Gemini Notebook Studio

LEARNING OBJECTIVES

- Match a chart type to the comparison or relationship you want a reader to see first.
- Recognise a misleading chart (distorted axis, inconsistent scale) and correct it.
- Recreate a chart from real data using the Chart-Type Decision Helper's recommendation.
- Generate a Studio infographic from verified sources and fact-check every claim in it.
- Select the right bibliometric or scientific-communication tool for a mapping or illustration task.

BEFORE YOU START

- Bring the findings or dataset you want to visualise. Even rough numbers or a summary table is enough.
- Decide who your audience is (academic/technical vs general) before choosing a chart type.
- If practising bibliometric mapping, export a reference list (BibTeX/RIS/CSV) from Module 1's Publish or Perish exercise to import into VOSviewer or Bibliometrix.

KEY CONCEPTS

Glossary of terms

Terms you will meet throughout this module, defined in plain language.

Co-citation network
A map showing which papers are frequently cited together, indicating related themes.

Bibliometric analysis
The quantitative study of publication and citation patterns.

Choropleth map
A map shading regions by a numeric value to show geographic patterns.

Data storytelling
Presenting data with a narrative and clear takeaway, not just raw numbers.

Trainer's Tip, Dr. Shahizan: Choose the chart type before you choose the colours. Decide what comparison or relationship you want the reader to see first, then pick a chart that shows exactly that. AI tools are fast at making pretty charts and slow at knowing what you are trying to say.

AI TOOLS FOR THIS MODULE

Tools for bibliometric mapping and scientific communication

TOOL	CATEGORY	WHAT IT DOES	WHERE TO ACCESS
VOSviewer	Bibliometric network	Constructs and visualises co-authorship, citation and keyword co-occurrence networks.	vosviewer.com
Bibliometrix	Bibliometric analysis	R-based toolkit for comprehensive bibliometric analysis and science mapping.	bibliometrix.org
Gephi	Network analysis	Open-source platform for visualising and analysing complex networks.	gephi.org
Open Knowledge Maps	Knowledge map	Creates visual overviews of research topics from scholarly literature.	openknowledgemaps.org
BioRender	Scientific illustration	Professional scientific illustrations for biology, medicine and life sciences.	biorender.com
Canva	Scientific figures & comms	Presentations, posters, infographics and other visual research communication.	canva.com

HOW TO USE

Step-by-step tool guides

STEP-BY-STEP GUIDE

Chart-Type Decision Helper

Describe what your data needs to show. The helper recommends suitable chart types and explains why, so you can brief an AI tool or design software with confidence.

01

Describe

State your goal — comparison, trend, part-to-whole, relationship, distribution, process, or geography.

02

Specify

Note the number of variables (1, 2, or 3+) and your audience (academic/technical or general).

03

Apply

Build the recommended chart type, keeping axes honest and units clearly labelled.

STEP-BY-STEP GUIDE

Gemini Notebook Studio (Infographics)

Gemini Notebook Studio can turn your uploaded sources into a polished infographic in seconds. The skill is not making the infographic — it is checking it.

01

Upload & generate

Upload 3–5 verified sources into Gemini Notebook, open Studio, and generate an infographic.

02

List every claim

Write down every number, statistic or claim shown in the infographic.

03

Fact-check

Check each claim against the uploaded sources: mark Correct, Wrong, or Not traceable — then fix or remove anything that fails.

HANDS-ON PRACTICE

All exercises, with model answers

Every exercise below matches the hands-on activity on the module's web page. Attempt each one yourself first, then compare against the model answer.

EXERCISE 4.1

~10 min

Chart-Type Decision Helper

Describe what your data needs to show. The helper recommends suitable chart types and explains why, so you can brief an AI tool or design software with confidence.

- [ ] State your goal: comparison between categories, trend over time, part-to-whole proportion, relationship between variables, distribution of values, process/workflow, or geographic pattern.
- [ ] Note the number of variables (1, 2, or 3+).
- [ ] Choose your audience: academic/technical or general.
- [ ] Run the recommendation and build the suggested chart type, checking axes and labels are honest before presenting it.

✓ WORKED EXAMPLE — GENERATED RECOMMENDATION

Sample input — Goal: Trend over time · Variables: 2 variables · Audience: General audience.

Recommended chart types: Line chart, which shows change or trend across a continuous time period; or an area chart, which emphasises cumulative volume or magnitude over time.

Tip: simplify labels, add a short takeaway title, and avoid dense technical notation for a general audience.

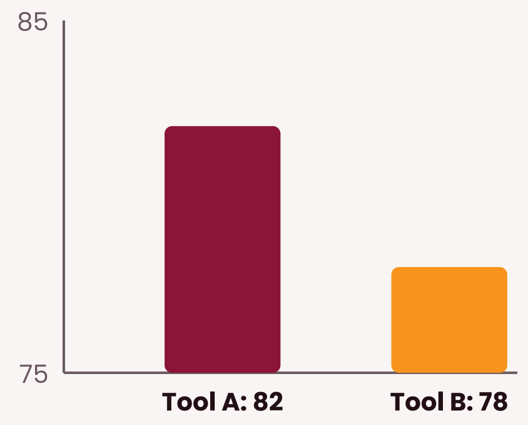
EXERCISE 4.2

~10 min

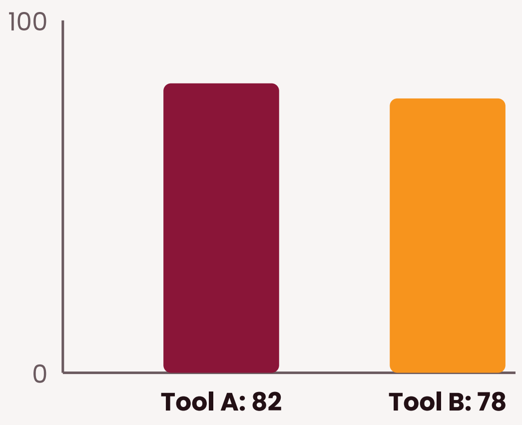
Spot the misleading chart

The same data can look completely different depending on how the axes are drawn. Both charts below plot the exact same two numbers, Tool A at 82 and Tool B at 78, a real difference of only 4 points.

- [ ] Does the y-axis start at zero? If not, is the break clearly marked and justified?
- [ ] Is the scale consistent across every bar, point or panel in the chart?
- [ ] Are the categories being compared actually comparable (same time period, same population, same unit)?
- [ ] Would the chart's headline takeaway change if you redrew it with an honest axis?



Distorted axis



Honest axis

WHY THIS MATTERS

- Distorted axis (left): the y-axis starts at 75, not 0. That makes Tool B's bar look less than half of Tool A's, when the real difference is only 4 points.
- Honest axis (right): the y-axis runs the full 0 to 100 scale. Tool A and Tool B now look as close as they actually are.
- Apply this to your own charts: run every chart you plan to present, including one an AI tool generated for you, through the four questions above before you publish or present it.

EXERCISE 4.3

~15 min

Recreate a chart from your own data

Use the Chart-Type Decision Helper with real numbers instead of a hypothetical scenario.

- [ ] Pull a small dataset you already have, for example the number of matching papers per year from your Module 1 Publish or Perish search, or your own study's results.
- [ ] Fill in the table below with your actual numbers.
- [ ] Decide your goal (comparison, trend, proportion, relationship, distribution) and run it through the Chart-Type Decision Helper.
- [ ] Build the recommended chart in a tool of your choice (Excel, Canva, or a bibliometric tool), and check it against the four questions from Exercise 4.2.

YEAR	COUNT
2021	
2022	
2023	
2024	

✓ MODEL ANSWER — SAMPLE FILLED DATA & CHART CHOICE

Sample data: 2021 → 12 papers; 2022 → 19 papers; 2023 → 31 papers; 2024 → 40 papers (partial year).

Goal: trend over time, 2 variables (year, count), general audience.
Recommended chart: a line chart showing steady year-on-year growth.
Four-question check: y-axis starts at 0 (pass); scale is consistent across all four years (pass); years are directly comparable, same search string each time (pass); 2024 is a partial year, so the chart notes "2024 (partial)" on the label to avoid implying a plateau or decline that isn't real.

EXERCISE 4.4

~15 min

Generate and fact-check a Studio infographic

Gemini Notebook Studio can turn your uploaded sources into a polished infographic in seconds. The exercise is not making the infographic, it is checking it.

- [ ] Upload 3 to 5 verified sources on your topic into Gemini Notebook, then open Studio and generate an infographic.
- [ ] List every number, statistic or claim shown in the infographic.
- [ ] Check each one against the actual uploaded sources. Mark it Correct, Wrong or Not traceable to a source.
- [ ] Fix or remove anything Wrong or Not traceable before you would consider using the infographic anywhere real.

CLAIM OR NUMBER IN THE INFOGRAPHIC	CHECK RESULT
"68% of respondents reported improved efficiency"	Correct, matches Table 2 in Source A

✓ MODEL ANSWER — SAMPLE COMPLETED FACT-CHECK

- "68% of respondents reported improved efficiency" — Correct, matches Table 2 in Source A.
- "Adoption has tripled since 2020" — Wrong: Source B actually reports a doubling (from 18% to 36%), not tripling — the infographic overstated the figure.
- "Most researchers now use AI tools daily" — Not traceable: no uploaded source reports a "daily use" statistic; this line was removed before using the infographic.

WHY USE AI HERE

Traditional vs AI-assisted approach

Task	Traditional Approach	AI-Assisted Approach
Choosing a chart type	Guess based on habit or what's familiar	Chart-Type Decision Helper recommends types based on your actual goal
Mapping citation networks	Manually tally co-citations and co-authorships	VOSviewer or CiteSpace generate the full network automatically
Getting a topic overview	Read through papers and sketch a summary by hand	Open Knowledge Maps generates a visual topic overview from the literature
Producing a polished figure	Rebuild formatting manually in a general design tool	Canva or BioRender offer research-ready templates for posters and figures

The trade-off: AI speeds up chart selection and network mapping; you still own the accuracy and honesty of the final visual.

COMMON PITFALLS

Things to verify before you trust the output

- [ ] Letting the y-axis start above zero without a clearly marked, justified break — the single most common way a chart misleads.
- [ ] Publishing an AI-generated infographic without checking every number in it against the actual uploaded sources.
- [ ] Choosing a chart type based on habit rather than the specific comparison, trend or relationship you want the reader to see.
- [ ] Comparing categories that are not actually comparable (different time periods, populations or units) in the same chart.
- [ ] Treating a partial-year or provisional data point the same as a complete one, implying a trend that isn't real yet.

Key Takeaways

- Match the chart type to your goal (comparison, trend, proportion, relationship, distribution, process or geography), rather than choosing a chart first and forcing the data to fit it.
- Bibliometric tools (VOSviewer, CiteSpace, Gephi, Bibliometrix) reveal patterns across a whole literature, not a single study.
- Keep axes honest and labels clear. A chart should clarify data, not distort it.
- Studio prompts can generate a source-grounded visual directly from your verified papers — but you still fact-check every claim.

What's next: Have your findings clearly visualised? Module 5 helps you weave them into a well-structured, properly cited piece of academic writing.

FOR YOUR NOTES

Session Notes

FOR YOUR NOTES

Session Notes (continued)